

The Pile Game

Week 1 · Grades 2–3

Name: _____

Mathematicians call this game **Bulgarian solitaire**. A real theorem is hiding inside it, and today you are going to find it with counters, paper and your table-mates.

The rule

Take one counter from **every** pile. Put those counters together: they make a **new pile**. A pile that had only one counter is **gone**. That is one **move**. Then move again, and again.

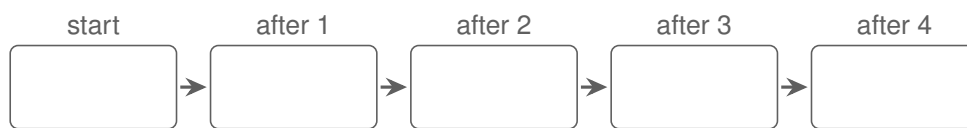


The dark counters are the ones being taken; after the move they form the dark pile. In the first picture two piles of one disappear. In the second picture no pile disappears, and the new pile is the biggest.

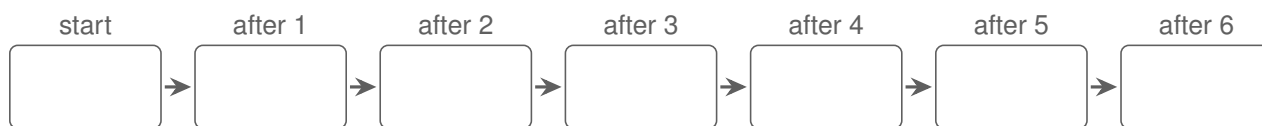
How we write a position: the pile sizes, biggest first, like **3 1 1**. Where the piles sit on the table does not matter—only their sizes do.

Warm-up: six counters

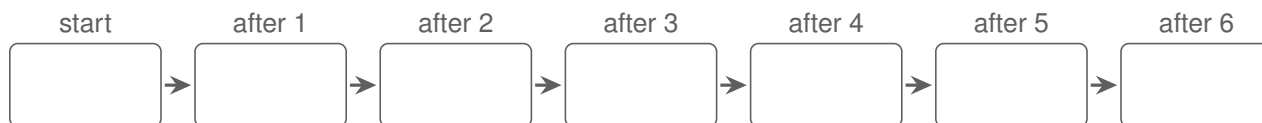
A. Start with all six counters in one pile: **6**. Play until the piles stop changing. Write every position.



B. Now start at **4 1 1**.



C. Start any way you like—a different start from everyone else at your table.



When the piles stay the same after a move, we say the game is **stuck**. Where did each game get stuck? What do you notice?

The map of six

There are exactly **11** ways to split 6 counters into piles. Here they all are. For each one, make it with your counters, do **one** move, and write what you get. Share the work: each person at the table takes a few.

	start		after one move		start		after one move
	6	→			3 1 1 1	→	
	5 1	→			2 2 2	→	
	4 2	→			2 2 1 1	→	
	4 1 1	→			2 1 1 1 1	→	
	3 3	→			1 1 1 1 1 1	→	
	3 2 1	→					

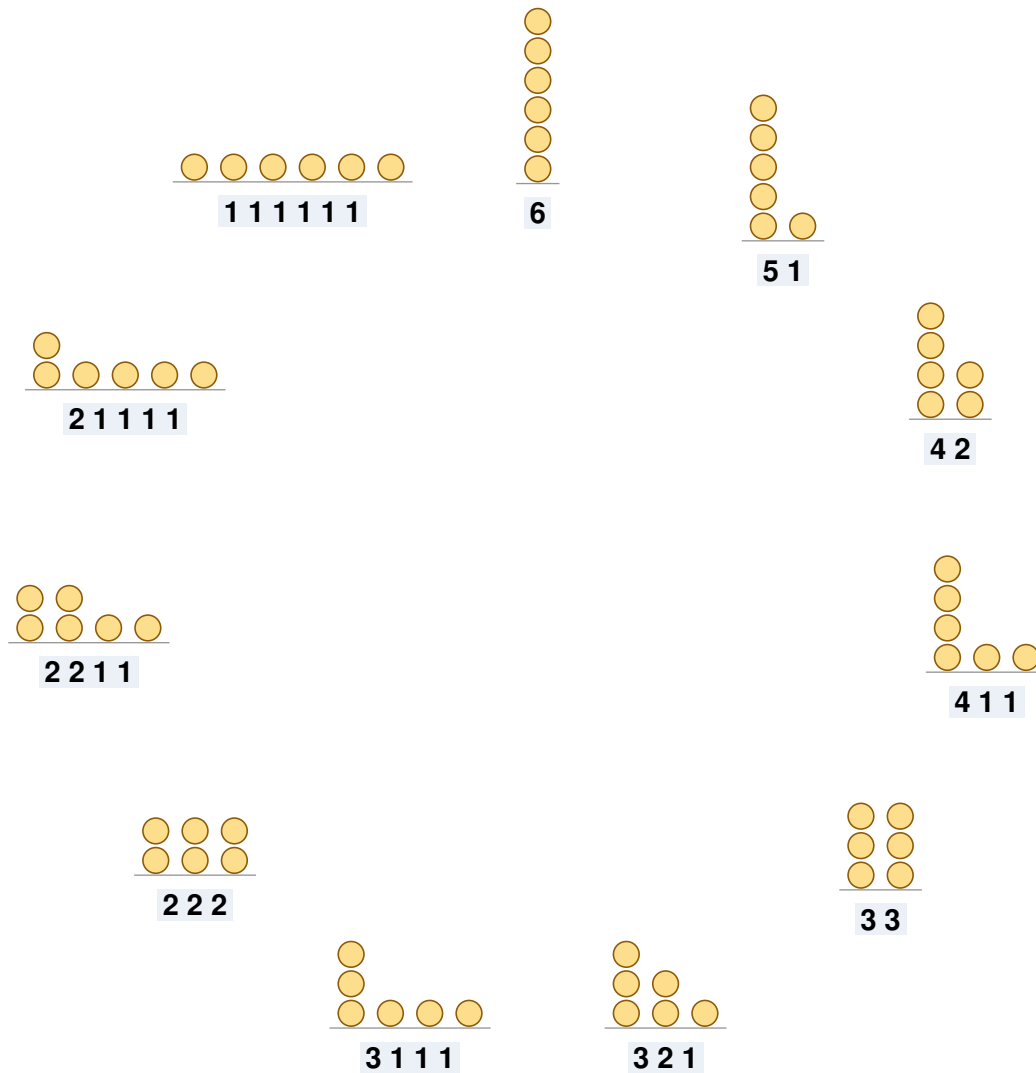
Check: every answer should be one of the 11 positions in the list. Is it?

Think: why are there no other ways to split 6 counters? (Try to find a twelfth one. What goes wrong?)

Now turn the page and draw the map.

Draw the map

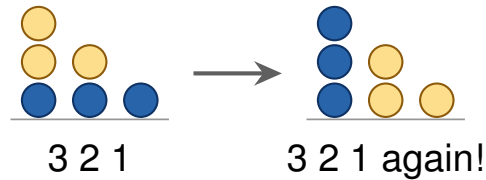
All 11 positions are arranged in a circle. Draw an arrow from each position to the position it goes to after one move. (If a position goes to itself, draw a little loop.)



1. Where do all the roads end? _____
2. Which start has the **longest** road to the end? _____ How many moves? _____
3. Some positions have **no arrow pointing at them**: nothing leads there, so you can only *start* there. Circle them on the map. How many are there? _____

Why does the staircase stay stuck?

3 2 1 is a **staircase**: each pile is one shorter than the one before. Lay the piles out as columns, side by side, like in the picture. Taking one counter from every pile is the same as taking the whole **bottom row**.



The bottom row has 3 counters, because there are 3 piles. It stands up and becomes the new pile of 3. The old piles shrink to 2, 1, and nothing. So the move gives **3 2 1** again. Stuck!

1. Explain to your table why **4 3 2 1** is stuck too—*without* playing it. Then play it to check.

2. These are the **staircase numbers**: the numbers of counters in a staircase.

1, 3, 6, 10, _____, _____, _____

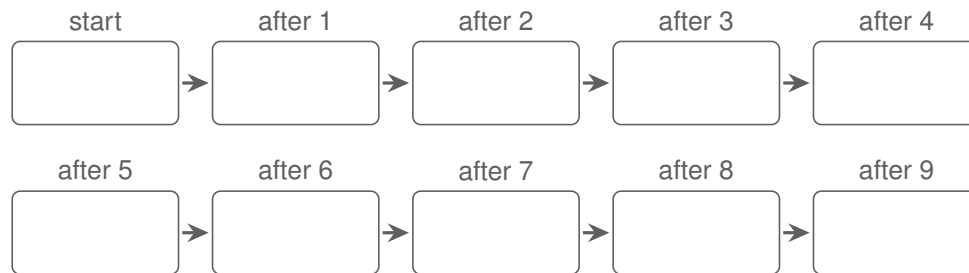
How do you get from each staircase number to the next?

3. Is 8 a staircase number? _____ Try to build a staircase with 8 counters. What goes wrong?

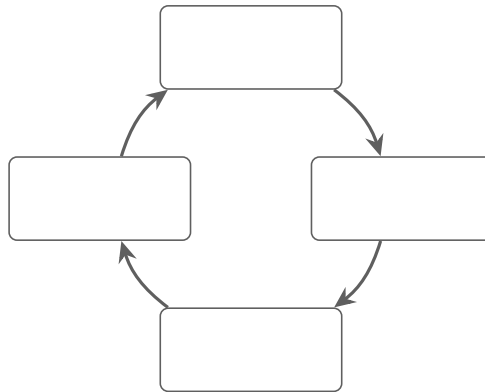
4. Mathematicians have proved that **only staircases are stuck positions**. So can a game with 8 counters ever get stuck? Explain why or why not.

Eight counters

Eight counters can never get stuck. So what *does* happen? Pick any start, play, and write every position. Keep going until you are sure what is happening.



Sooner or later a position comes back. From then on the game goes round and round: a **loop**. Write the positions of your loop in the boxes (you may not need all four).

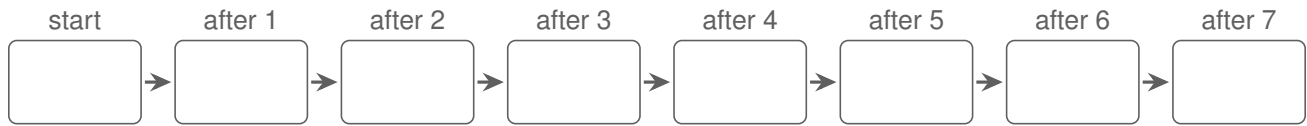


1. How many positions are in your loop? _____
2. Compare with your table. Did everyone find the **same** loop? _____
3. There are **two different loops** for 8 counters. Find the other one. (Hint: try starting at **5 3** or at **2 2 2 2**.)

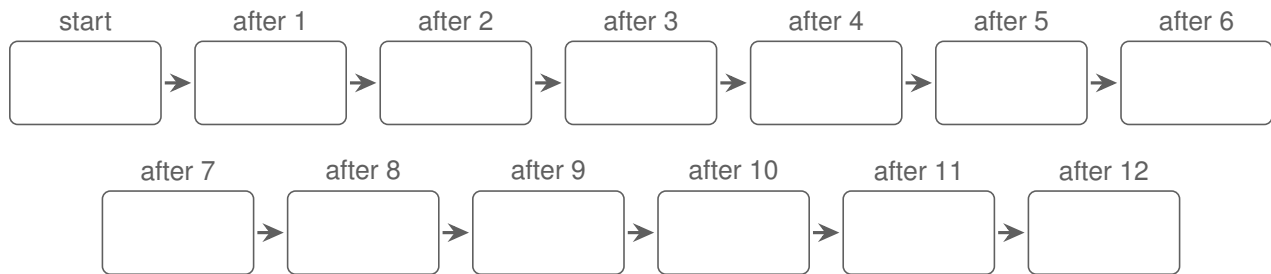
4. Every loop position looks like the staircase **3 2 1** with two extra counters added. Where do the extra counters sit? Make the loop positions with your counters and look.

Challenges

1. Seven counters. Stuck, or a loop? How many positions are in the loop?



2. The longest road. With 10 counters every start ends at **4 3 2 1**—that is what happened when we were the counters. Some starts get there fast: **10** takes 6 moves. The record is **12 moves**. Find a start that takes at least 8 moves. Can you find one that takes 12? (Hint: use five or six piles.)



3. Nobody can get there. On your map of six, three positions have no arrow pointing at them. Write them here. What do they have in common? Then find such a position for 7 counters and check that nothing leads to it.

4. Make your own map. Draw the whole map for 4 counters (there are 5 positions) or for 5 counters (there are 7 positions). Does every road end at the same place?